

REMARKS ON SIX-TANGENTIAL CONICS TO SMOOTH PLANE SEXTICS

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ABSTRACT.

Let k be a positive integer. We consider the complete graph with vertex set $V_k = \{1, \dots, k\}$ and edge set

$$E_k := \{[i, j] \mid i, j \in V_k, i < j\}.$$

Let m be a positive integer, and we put

$$M := \{-m, -m+1, \dots, m\}.$$

Let $\mathcal{W}_k$ be the set of weights $w: E_k \rightarrow M$, that is, $\mathcal{W}_k$ is the set of weighted complete graphs on V_k with weights in M . We let the symmetric group $\mathfrak{S}_k$ act on V_k from the right. Then $\mathcal{S}_k$ acts on $\mathcal{W}_k$ from the left. We define Sw_k to be the multiplicative group of all mappings $s: V_k \rightarrow \{\pm 1\}$, which acts on $\mathcal{W}_k$ by switching, that is,

$$(sw)([i, j]) = s(i)s(j)w([i, j])$$

for $s \in \text{Sw}_k$ and $[i, j] \in E_k$. Then the semidirect product

$$G_k := \text{Sw}_k \rtimes \mathfrak{S}_k$$

acts on $\mathcal{W}_k$ from the left. The purpose of this section is to describe an efficient algorithm for obtaining the list of orbits of the action of G_k on $\mathcal{W}_k$.

First, we introduce a total order on the set E_k of edges by

$$[i, j] < [i', j'] \iff j < j' \text{ or } (j = j' \text{ and } i < i').$$

Then we express elements w of $\mathcal{W}_k$ by arranging the values $w([i, j])$ in this order:

$$w = (w([1, 2]), w([1, 3]), w([2, 3]), w([1, 4]), \dots).$$

Let $\text{ab}(w)$ denote the vector obtained from w by taking the absolute value of each component. We put $\text{ab}(M) := \{0, 1, \dots, m\}$. Then we have a mapping

$$(0.1) \quad \text{ab}(\cdot) : \mathcal{W}_k \longrightarrow \text{ab}(\mathcal{W}_k) := (\text{ab}(M))^{E_k}, \quad w \longmapsto \text{ab}(w).$$

Definition 0.1. Suppose that a group G acts on a set X , and that X is equipped with a total order $<_X$. An element $x \in X$ is $(G, <_X)$ -*minimal* if x is minimal with respect to $<_X$ in the orbit $G \cdot x$.

Notice that $\mathfrak{S}_k$ acts on $\text{ab}(\mathcal{W}_k)$, and the mapping (0.1) is $\mathfrak{S}_k$ -equivariant. Moreover, the action of Sw_k on $\mathcal{W}_k$ preserves each fiber of (0.1). We then define a total order $<_{\mathcal{W}}$ on $\mathcal{W}_k$ by

$$w <_{\mathcal{W}} w' \iff \text{ab}(w) <_{\text{lex}} \text{ab}(w') \quad \text{or} \quad (\text{ab}(w) = \text{ab}(w') \quad \text{and} \quad w <_{\text{lex}} w').$$

We obtain the list of G_k -orbits in $\mathcal{W}_k$ by enumerating all $(G_k, <_{\mathcal{W}})$ -minimal elements in $\mathcal{W}_k$.

We define the notion of G_k -quasi-minimality by the following:

$$w \text{ is } G_k\text{-quasi-minimal} \iff \begin{array}{l} \text{ab}(w) \text{ is } (\mathfrak{S}_k, <_{\text{lex}})\text{-minimal, and} \\ w \text{ is } (\text{Sw}_k, <_{\text{lex}})\text{-minimal in } \text{ab}^{-1}(\text{ab}(w)). \end{array}$$

It is obvious that, if $w \in \mathcal{W}_k$ is $(G_k, <_{\mathcal{W}})$ -minimal then $w \in \mathcal{W}_k$ is G_k -quasi-minimal while the opposite implication does not necessarily hold. We first make the list of G_k -quasi-minimal elements, and then we choose from them $(G_k, <_{\mathcal{W}})$ -minimal elements.

To make the list of G_k -quasi-minimal elements, we use the induction on k . Suppose that $k' < k$. By taking the prefix of length $k'(k' - 1)/2$ of each $u \in \text{ab}(\mathcal{W}_k)$, we obtain a mapping

$$\text{ab}(\mathcal{W}_k) \longrightarrow \text{ab}(\mathcal{W}_{k'}), \quad u \longmapsto u|_{k'}.$$

It is obvious that, if $u \in \text{ab}(\mathcal{W}_k)$ is $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal, then $u|_{k'}$ must be $(\mathfrak{S}_{k'}, <_{\text{lex}})$ -minimal.

For an $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal element $u \in \text{ab}(\mathcal{W}_k)$,

$$\text{ab}^{-1}(u) \longrightarrow \text{ab}^{-1}(u|_{k'}), \quad w \longmapsto w|_{k'}.$$

It is obvious that, if $w \in \text{ab}^{-1}(u)$ is $(\text{Sw}_k, <_{\text{lex}})$ -minimal, then $w|_{k'}$ must be $(\text{Sw}_{k'}, <_{\text{lex}})$ -minimal. More strongly, $w \in \text{ab}^{-1}(u)$ is $(\text{Sw}_k, <_{\text{lex}})$ -minimal only if $w|_{k'}$ is $(\text{Sw}_{k'}, <_{\text{lex}})$ -minimal and the first non-zero element of the suffix $w - w|_{k'}$ is negative. (The opposite implication does not necessarily hold.) We say that w is $(\mathfrak{S}_k, <_{\text{lex}})$ -quasi-minimal if, for any $k' < k$, $w|_{k'}$ is $(\mathfrak{S}_{k'}, <_{\text{lex}})$ -quasi-minimal and the first non-zero element of the suffix $w - w|_{k'}$ is negative.

We enumerate all $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal elements and then, for each $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal element $u \in \text{ab}(\mathcal{W}_k)$, we enumerate all $(\text{Sw}_k, <_{\text{lex}})$ -quasi-minimal elements in $\text{ab}^{-1}(u)$.

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